TRANSIT DATA CONFERENCE 2026

Toronto, Canada | June 23-25, 2026

The Transit Data Challenge

Official Rules, Guidelines & FAQ

Advancing the Future of Public Transit Through Data Innovation

1. Introduction & Purpose

The Transit Data Challenge invites student teams from Canadian universities to tackle real-world challenges in public transit data. Participants will develop innovative solutions at the intersection of transportation, data science, and public policy — showcasing how advanced analytics, artificial intelligence, and modern data infrastructure can make transit systems smarter, more equitable, and more responsive to the communities they serve.

This competition is designed to bridge academia and industry, giving students hands-on experience with transit data problems while contributing meaningful work to the broader practitioner and research community attending the Transit Data Symposium 2026.

Website	For full information, updates, and registration, visit: www.transitdata2026.ca

2. Transit Data Challenge Themes & Topics

Projects must address one or more of the following thematic areas:

- Advanced data analytics and artificial intelligence in public transit
- Applications using automated data to enhance strategic or service planning, performance measurement, fare policy, market research, and equity analysis
- Applications using automated data to assist in operations control, incident management, and service restoration

- Use of automated data to understand demand, customer behaviour, normal travel patterns, and response to service disruptions
- Methodological challenges in using automated data
- New technologies and approaches for data collection
- Visualization tools for transit data communication
- Organizational challenges related to data management and governance
- Transit automated data specifications and their practical use (e.g., GTFS, TIDES, TODS)

Teams are encouraged to be creative and cross-disciplinary. Projects may combine multiple themes and are expected to demonstrate both technical rigour and practical relevance to transit agencies, policy-makers, or transit users.

3. Eligibility & Team Composition

3.1 Who May Participate

The Transit Data Challenge is open to currently enrolled students at accredited Canadian universities and colleges (undergraduate, graduate, or doctoral level). Students from any discipline are welcome, including but not limited to: computer science, data science, engineering, urban planning, public policy, geography, statistics, and business.

3.2 Team Structure

Each team must be composed of:

Role	Requirements
Student Members	Minimum 1, maximum 3. Must be enrolled at a Canadian university or college.
Faculty/Industry Mentor (optional but strongly recommended)	Exactly 1 mentor affiliated with a Canadian university. Mentors provide guidance only and do not perform hands-on development work.
Maximum Team Size	4 members total (3 students + 1 optional mentor).

Note	Students may only participate in one team. A mentor may support more than one team, but this must be declared at registration, and the same mentor cannot be the sole mentor across more than two teams. The personnel involved in organizing the Transit Data Challenge and Transit Data Symposium 2026 event will not participate as mentors.

> Members of the scientific committee, who are professors, are allowed to serve as mentors.

4. Data Requirements & Privacy Policy

4.1 Privacy-First Principle

Transit Data Challenge is committed to responsible data use. All projects must adhere strictly to a Privacy-First Principle. This is a non-negotiable requirement and violations will result in immediate disqualification.

4.2 Permitted Data Sources

Teams must use only the following types of data:

- Synthetic data: artificially generated datasets that replicate realistic transit patterns without containing any real person's information.
- Open/public data: publicly available datasets released under open data licences (e.g., open government data portals, OpenStreetMap, GTFS feeds published by transit agencies, Statistics Canada open data).

4.3 Prohibited Data

The following are strictly prohibited:

- Any dataset containing real Personally Identifiable Information (PII), including but not limited to: names, addresses, phone numbers, email addresses, payment card numbers, travel history linked to individuals, facial recognition data, or device identifiers.
- Scraped or harvested data from platforms where collection violates terms of service.
- Proprietary transit agency operational data obtained through non-public agreements.
- Any data that, even if publicly available, could be combined to re-identify individuals (data linkage attacks).

> **IMPORTANT**
> Teams are responsible for verifying the provenance and licence of all datasets used. The organizing committee reserves the right to request data documentation at any time. Non-compliance will result in disqualification.

4.4 Data Documentation

At submission, teams must provide a Data Sources Statement listing every dataset used, its origin, the licence under which it is used, and confirmation that no PII is present.

5. Competition Phases & Timeline

Phase	Date	Description
Registration Opens	**March 15, 2026**	Teams register at www.transitdata2026.ca. All team members must be listed. (click here)
Submission Deadline	**May 30, 2026**	Full project submission due (see Section 6). No late submissions accepted.
Shortlist Announced	**June 5, 2026**	Three finalist teams selected and notified by the organizing committee.
Final Presentations	**June 25, 2026**	Finalist teams present live at Transit Data Symposium 2026 in Toronto. Winners announced at the closing ceremony.
Prototype	**June 25 to July 31, 2026**	The period after TD2026 during which selected work will be displayed online.

Timezone	All deadlines are at 11:59 PM Eastern Time (ET) on the specified date.

6. Phase 1 — Registration & Project Submission

6.1 Registration

Teams must register through the official Transit Data 2026 website (www.transitdata2026.ca). During registration, teams must provide:

1. Team name and list of all members (full name, university, programme, and year of study for each student; name, affiliation, and role for the mentor).
2. Project title and abstract (250–400 words) describing the problem addressed, proposed methodology, and anticipated impact.
3. Thematic area(s) targeted (from Section 2).
4. Confirmation that the Privacy-First Principle (Section 4) will be adhered to.

6.2 Final Project Submission

By the submission deadline (May 30, 2026), teams must submit the following package through the online portal:

- Technical Report (PDF, maximum 15 pages excluding references and appendices): describing the problem statement, methodology, data sources, results, limitations, and conclusions.
- Data Sources Statement (see Section 4.4).
- Working Prototype: a live, accessible link to the interactive tool, web application, or mobile app (see Section 7). The prototype must be functional at the time of submission and remain accessible until July 31, 2026.
- Short Video Walkthrough (maximum 3 minutes): a screen-recorded demonstration of the prototype explaining key features and findings. The video upload will be detailed in an official email invitation for the finalists.
- Source Code Repository: a link to a public GitHub/GitLab repository with a README describing how to run the project locally.

Incomplete submissions will not be reviewed. All materials must be submitted in English or French.

7. Final Deliverable Requirements

7.1 Interactive Tool / Application

The centrepiece of each team's submission must be an interactive tool with a graphical user interface. This deliverable will be demonstrated live during the conference presentation. Acceptable formats include:

- Web application or web tool (accessible via standard browser, no installation required)
- Mobile application (iOS or Android; must be testable without App Store/Play Store accounts, e.g., via TestFlight or APK)
- Desktop application with graphical interface (must run on Windows or macOS)

The tool must meet the following standards:

- Functionality: demonstrates the core analytical or visualization capability described in the technical report.
- Usability: navigable by a non-technical user; key features must be self-evident or accompanied by in-app guidance.
- Graphical Interface: must include meaningful data visualizations (maps, charts, dashboards, or similar).
- Accessibility: wherever possible, follow WCAG 2.1 Level AA accessibility guidelines.
- Performance: must load and respond within reasonable timeframes during a live demonstration on conference Wi-Fi.

Tip	Web tools are strongly recommended for ease of demonstration at the conference. Ensure your application can run without requiring specialized hardware or a local development environment.

7.2 Presentation at Transit Data Symposium 2026

Each of the three finalist teams will receive a dedicated time slot during the Transit Data Symposium 2026 on June 25, 2026. The presentation format is:

- Duration: 10 minutes presentation + 5 minutes Q&A.
- Format: live demonstration of the interactive tool, supported by slides (optional).
- Presenters: all student team members are expected to participate in the presentation. The mentor may be present but should not lead the presentation.
- Technical setup: teams will be provided with a projector/screen and a reliable internet connection. Teams are responsible for bringing their own laptops. Technical support will be available 30 minutes before the first session.

Presentations will be evaluated by the judging committee and open to the full conference audience for the popular vote component.

8. Selection Process & Judging Criteria

8.1 Shortlisting (Phase 1 Review)

All complete submissions will be reviewed by a panel of at least three independent reviewers drawn from the organizing committee. The three highest-scoring teams will be selected as finalists. Reviewers will assess submissions on the criteria below. In the event of a tie, an additional reviewer will be consulted.

8.2 Evaluation Criteria

Criterion	Description	Weight
Innovation & Originality	Novelty of the approach, creativity in problem framing, and originality of the solution.	**20%**
Technical Merit	Soundness of methodology, appropriate use of data analytics/AI, robustness of implementation.	**25%**
Relevance to Transit	Alignment with Transit Data Challenge themes and practical applicability to transit agencies or policy-makers.	**20%**
Interactive Tool Quality	Quality of the graphical interface, usability, and effectiveness of visualizations.	**20%**

Presentation & Communication	Clarity of the live presentation, ability to answer questions, and quality of the technical report.	**10%**
Privacy & Ethics Compliance	Demonstrated adherence to the Privacy-First Principle and responsible data practices.	**5%**

8.3 Final Winner Selection

The winner will be determined by a blended scoring process involving three components:

- Scholar & Researcher Vote: a panel of academic reviewers evaluating technical quality and innovation.
- Industry Partner Vote: representatives from transit agencies, technology companies, and public sector partners assessing practical applicability.
- Popular Vote: conference attendees vote for their favourite presentation via a designated digital voting mechanism. All registered attendees of the Transit Data Symposium 2026 may cast one vote.

The exact weighting between the three voting components will be announced at the conference. All three finalist teams will be recognized in the conference proceedings regardless of placement.

The personnel involved in the Transit Data Symposium 2026 Organizing Committee event will not participate in the Scholar & Researcher Vote.

9. Prizes

Prizes are to be announced. The organizing committee is actively engaging industry and government partners to secure meaningful awards that recognize the effort and talent of participating teams. Prize details will be published on www.transitdata2026.ca as they are confirmed.

At a minimum, all finalist teams will receive:

- Complimentary registration to the Transit Data Symposium 2026 for all student team members.
- Formal recognition in the Transit Data Symposium 2026 programme and proceedings.
- Certificate of participation signed by the Transit Data Symposium 2026 organizing committee.
- Opportunity to network with leading transit practitioners, researchers, and technology partners.

10. Code of Conduct & Integrity

All participants are expected to uphold the highest standards of academic and professional integrity. The following rules apply:

- All work submitted must be original and created by the student members of the team during the competition period. Use of pre-existing personal or open-source code libraries is permitted, but must be disclosed.
- Teams must not misrepresent the capabilities of their prototype during the presentation.
- Plagiarism, falsification of results, or misrepresentation of data sources will result in immediate disqualification and may be reported to the participants' academic institutions.
- Respectful and inclusive conduct is required at all times, both in online interactions and at the conference venue. Harassment, discrimination, or intimidation of any kind will not be tolerated.
- Teams must not use the Transit Data Challenge to promote commercial products or services in which they have a financial interest.

By participating, all team members and mentors agree to abide by this Code of Conduct and all rules outlined in this document.

11. Intellectual Property

Teams retain ownership of their intellectual property. By submitting to the Transit Data Challenge, teams grant the organizing committee a non-exclusive, royalty-free licence to:

- Feature their project (including name, abstract, screenshots, and video) in Transit Data Symposium 2026 communications, publications, and promotional materials.
- Include a summary of their work in the conference proceedings.

Transit Data Symposium 2026 will not claim ownership of any submitted work or commercialize team creations without explicit written consent.

12. Contact & Organizing Committee

For questions, registration issues, or clarifications not addressed in this document or the FAQ below, please contact the Transit Data Challenge team via the official website at www.transitdata2026.ca or tal@utoronto.ca

Decisions of the organizing committee and the judging panel are final. The organizing committee reserves the right to amend these rules at any time; any changes will be communicated promptly via the official website and to all registered teams by email.

Frequently Asked Questions (FAQ)

The following questions and answers are provided to help teams understand the Transit Data Challenge process. This section will be updated as additional questions are received. Always check www.transitdata2026.ca for the latest version.

Registration & Eligibility

Q: Can students from different universities form a team?

A: Yes. Teams may be composed of students from different Canadian universities or colleges, provided each member individually meets the eligibility criteria. The team must have a single designated lead for communication purposes.

Q: Can a graduate student and an undergraduate student be on the same team?

A: Yes. Teams may include any mix of undergraduate, graduate, or doctoral students, as long as each is currently enrolled at an accredited Canadian university or college.

Q: Who qualifies as a mentor?

A: A mentor can be either a faculty member or researcher affiliated with a Canadian university or college. Mentors may be declared at registration and provide their name, title, and institutional affiliation. Although not mandatory, we strongly recommend inviting a mentor.

Q: Can a professor or industry professional be a student AND a mentor?

A: No. Mentors and students are distinct roles. A mentor cannot also be listed as a student team member.

Q: Is there a registration fee?

A: There is no registration fee for the Transit Data Challenge itself. Finalist teams will receive complimentary conference registration. Non-finalist teams wishing to attend Transit Data Symposium 2026 will need to register for the conference separately.

Data & Privacy

Q: Where can we find transit open data to use in our project?

A: Numerous sources are available, including: open.canada.ca (federal open data portal), transit agency GTFS feeds (many Canadian agencies publish these publicly), Statistics Canada open data (statcan.gc.ca), OpenStreetMap, provincial open data portals (e.g., Ontario Data Catalogue), and the Mobility Data Specification (MDS) repositories. A curated list of recommended sources will be posted on www.transitdata2026.ca.

Q: Can we use real transit smartcard (AFC) data if it has been anonymized?

A: Only if the data has been fully de-identified and is available publicly under an open data licence, or if its use has been authorized in writing by the relevant transit agency. "Anonymized" data that retains fields enabling re-identification (e.g., origin-destination pairs at fine spatial/temporal resolution) is still considered PII under these rules. When in doubt, use synthetic data instead.

Q: What is GTFS and where do we start?

A: GTFS (General Transit Feed Specification) is a standard format for public transit schedules and geographic information. TIDES (Transit ITS Data Exchange Specification) and TODS (Transit Operational Data Standard) are related standards for operational and automated data.

Q: Can we generate synthetic data based on a real city's transit network?

A: Yes. You may use the public structural information about a real transit network (e.g., routes, stops, and schedules from a public GTFS feed) as the basis for synthetic ridership or demand data. The key requirement is that no synthetic records should represent or be traceable to real individuals.

Project Submission

Q: Do we need to have a fully polished product by the submission deadline?

A: Your prototype must be functional and demonstrate your core concept clearly — it does not need to be a production-ready commercial product. Judges understand this is a student competition. Focus on demonstrating the core idea compellingly and ensuring the tool works reliably during the live demonstration.

Q: What technologies can we use to build our tool?

A: There are no restrictions on programming languages, frameworks, or libraries. Commonly used tools for transit data projects include Python (pandas, geopandas, scikit-learn, dash/plotly), R (shiny), JavaScript/TypeScript (React, Leaflet, Mapbox), and geospatial platforms (QGIS, Kepler.gl). The choice of technology is entirely up to your team.

Q: Our web app requires a backend server — how should we host it for the submission and presentation?

A: You are responsible for hosting your application for the duration of the competition (from submission to July 31, 2026). Free-tier cloud services (e.g., Heroku, Railway, Render, Vercel, Netlify) are sufficient for most projects. Ensure your application remains accessible without a local development environment. Include setup instructions in your README in case live access is unavailable during judging.

Q: Can we update our prototype after the submission deadline?

A: Minor bug fixes and performance improvements may be made after the submission deadline and before the conference, provided the core functionality and scope of the project do not change. Significant feature additions after the deadline are not permitted. Any updates must be disclosed to the organizing committee.

Q: In what language should we submit our materials?

A: All materials (technical report, video, README, application interface) must be submitted in English or French. Bilingual submissions are welcome. Presentation will be in English.

Presentations & Conference

Q: Do all team members need to travel to Toronto for the presentation?

A: All student team members are expected to present in person at Transit Data Symposium 2026 in Toronto on June 25, 2026. If exceptional circumstances prevent in-person attendance (e.g., serious illness), the organizing committee should be notified as early as possible and will assess alternatives on a case-by-case basis. Remote-only presentations are not the default option. The team members can also choose one student to serve as their representative.

Q: Will travel or accommodation support be provided to finalists?

A: Travel and accommodation arrangements are the responsibility of the teams. Some support may be made available depending on sponsor funding — watch www.transitdata2026.ca for announcements. Teams are encouraged to explore travel grants through their institutions.

Q: How does the popular vote work?

A: Conference attendees will be able to vote for their favourite finalist team's presentation through a digital voting mechanism (details to be announced). Every registered Transit Data Symposium 2026 attendee may cast one vote during the designated voting period, which will occur after all three presentations are complete. The popular vote contributes to the overall winner determination alongside the scholar and industry partner votes.

Q: Can non-finalist teams still attend Transit Data Symposium 2026?

A: Yes. Non-finalist teams are welcome to register for Transit Data Symposium 2026 through the standard registration process at www.transitdata2026.ca. All registered participants will be recognized in the conference programme.

Prizes & Recognition

Q: When will prize details be announced?

A: Prize details are to be determined and will be announced on www.transitdata2026.ca as soon as they are confirmed. The organizing committee is actively working with industry and government partners to secure meaningful awards.

Q: If we win, who owns the solution — can we commercialize it?

A: Teams retain full intellectual property rights over their work (see Section 11). You are free to commercialize, publish, or further develop your solution. Transit Data symposium 2026 only requests a non-exclusive licence for promotional and archival purposes.

General

Q: What if our question is not answered here?

A: Please submit your question through the contact tal@utoronto.ca. The FAQ will be updated regularly to reflect common inquiries. You can also follow Transit Analytics Lab (TAL) on social media for announcements and updates.

Q: Can we start building our project before registration opens?

A: Teams may explore ideas, conduct background research, and learn relevant tools before registration opens. However, the competitive project — particularly any proprietary datasets, application code, or analysis — should begin in earnest after registration so the process is fair to all teams.

These rules are subject to change. All updates will be communicated to registered teams and posted on the official website.